

AP Statistics FRQ Practice Exam 01

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AP Statistics FRQ Practice Exam 1

APSTAT-FRQ1-001

Section: FRQ

Type: frq

Primary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Secondary Unit: None

FRQ Type: Formulate Questions and Collect Data

Prompt:

A regional emergency-dispatch center employs 240 dispatchers. The center is considering a compact-alert screen intended to reduce the time needed to assign an ambulance unit correctly. The dispatchers are distributed as shown.

Shift	Dispatchers	Planned random sample
Day	120	24
Evening	90	18
Overnight	30	6
Total	240	48

The director will select a stratified random sample of 48 dispatchers, using shift as the strata and the proportional sample sizes shown. Each selected dispatcher will complete the same set of 12 realistic dispatch simulations. The response variable will be the median number of seconds the dispatcher takes to make a correct assignment.

- (a) Identify the population and the sample in this study.
- (b) Explain one reason the proposed stratified random sample could produce a more precise estimate of the population mean response time than a simple random sample of 48 dispatchers, assuming response times tend to differ by shift.
- (c) State a statistical question that could be answered by an experiment comparing the compact-alert screen with the current screen. Identify the explanatory variable and the response variable.
- (d) The 48 sampled dispatchers include 18 with less than 2 years of experience and 30 with at least 2 years of experience. Describe a randomized block design that uses experience level as the blocking variable. Explain the purpose of random assignment. If the compact-alert group has a meaningfully lower mean response time, state the scope of a conclusion supported by this design.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Identifies the population as all 240 dispatchers employed by the regional emergency-dispatch center.
- Point 2: Identifies the sample as the 48 dispatchers selected through the shift-stratified random sampling procedure.

(b) 2 points

- Point 1: Explains that stratifying guarantees representation from the day, evening, and overnight shifts in proportions matching the center's workforce.
- Point 2: Connects possible within-shift similarity and between-shift differences in response time to reduced sampling variability, compared with an SRS of the same size.

(c) 2 points

- Point 1: States a comparative statistical question about whether the mean or typical correct-assignment time differs between the compact-alert and current screens for the center's dispatchers.
- Point 2: Identifies screen type as the explanatory variable and median correct-assignment time, in seconds across the 12 simulations, as the response variable.

(d) 4 points

- Point 1: Separates the sampled dispatchers into the two experience blocks: less than 2 years and at least 2 years.
- Point 2: Randomly assigns approximately half of the dispatchers within each experience block to the compact-alert screen and the remainder to the current screen, then applies the same simulation protocol to both groups.
- Point 3: Explains that random assignment tends to balance lurking variables between screen groups and permits a cause-and-effect conclusion about screen type; blocking separately controls variation associated with experience.
- Point 4: States that, because the dispatchers were randomly sampled and screen type was randomly assigned, a causal conclusion may be generalized to the 240 dispatchers at this center, but not automatically to dispatchers at other centers.

Model Guidance:

The target population is the center's full workforce of 240, whereas the sample consists only of the 48 selected dispatchers. Proportional stratification prevents a small shift, especially the overnight shift, from being missed by chance. When people within a shift tend to have similar response times, estimating within each shift and combining the results can reduce sample-to-sample variability.

A suitable question is, "For dispatchers at this center, does use of the compact-alert screen change the mean median time to make a correct ambulance assignment compared with the current screen?" Screen type is explanatory, and median correct-assignment time is the response.

For the experiment, form the two stated experience blocks and randomize screen assignment separately inside each block. Use roughly equal treatment counts in each block and keep the simulations and other conditions the same. Random assignment supports causation by balancing other dispatcher characteristics; blocking prevents experience from obscuring the screen comparison. The random sample supports generalization to this center's dispatchers, while the design gives no statistical

basis for extending the conclusion to other centers.

Expected Points:

10

APSTAT-FRQ1-002

Section: FRQ

Type: frq

Primary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Analyze Data, One- or Two-Variable Data and Probability

Prompt:

A coastal field laboratory sends a technician on an 8-hour water-sampling shift with replacement filter cartridges. Let X be the number of cartridges that become unusable because of heavy sediment during a randomly selected shift in the autumn sampling season. Laboratory records support the following probability distribution.

x	0	1	2	3	4
$P(X = x)$	0.08	0.27	0.38	0.20	0.07

- (a) Verify that the table gives a valid probability distribution, and calculate $P(X \geq 3)$.
- (b) On paper, sketch a probability bar graph for X . Clearly label both axes and use a probability scale that shows all five bar heights.
- (c) Calculate the expected value of X . Interpret the expected value in context.
- (d) Calculate the standard deviation of X . Interpret the standard deviation in context.
- (e) Given that at least one cartridge becomes unusable during a shift, calculate the probability that at least three cartridges become unusable. Interpret the result in context.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Verifies that every probability is between 0 and 1 and that $0.08 + 0.27 + 0.38 + 0.20 + 0.07 = 1.00$.
- Point 2: Calculates $P(X \geq 3) = 0.20 + 0.07 = 0.27$.

(b) 2 points

- Point 1: Labels the horizontal axis as number of unusable cartridges, with values 0 through 4, and labels the vertical axis as probability with an appropriate scale beginning at 0.
- Point 2: Draws separate bars with heights 0.08, 0.27, 0.38, 0.20, and 0.07 at $x = 0, 1, 2, 3$, and 4, respectively.

(c) 2 points

- Point 1: Calculates $E(X) = \sum xP(X = x) = 0(0.08) + 1(0.27) + 2(0.38) + 3(0.20) + 4(0.07) = 1.91$ cartridges.
- Point 2: Interprets 1.91 as the long-run average number of cartridges that become unusable per 8-hour shift over many autumn sampling shifts, not as a possible count on one shift.

(d) 2 points

- Point 1: Calculates $\sigma^2 = \sqrt{\sum x^2P(X = x) - \mu^2} = \sqrt{(4.71 - 1.91^2)} \approx 1.03$ cartridges.
- Point 2: Interprets 1.03 cartridges as the typical distance between the number becoming unusable on a shift and the long-run mean of 1.91 cartridges.

(e) 2 points

- Point 1: Calculates $P(X \geq 3 | X \geq 1) = P(X \geq 3) \div P(X \geq 1) = 0.27 \div 0.92 \approx 0.293$.
- Point 2: Interprets the result: among autumn shifts on which at least one cartridge becomes unusable, about 29.3% are expected to have at least three unusable cartridges.

Model Guidance:

A discrete probability model is valid when its entries lie from 0 to 1 and sum to 1. The probability of three or more unusable cartridges is found by adding the entries for 3 and 4. A probability bar graph therefore has five bars located at the listed x-values, with the tallest bar, 0.38, at $x = 2$.

The distribution's balance point is $E(X) = 1.91$. This is a long-run average, so a noninteger value is reasonable even though X itself must be an integer. For spread, first find $E(X^2) = 4.71$; then $\sigma^2 = \sqrt{E(X^2) - E(X)^2} \approx 1.03$. The conditional probability in part (e) restricts the denominator to shifts with $X \geq 1$, whose probability is $1 - 0.08 = 0.92$. The event $X \geq 3$ lies entirely inside that restricted group, so the required ratio is $0.27/0.92 \approx 0.293$.

Expected Points:

10

APSTAT-FRQ1-003

Section: FRQ

Type: frq

Primary Unit: Unit 3, Inference for Categorical Data: Proportions

Secondary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

FRQ Type: Analyze Data, Statistical Inference (Confidence Interval or Significance Test)

Prompt:

A manufacturer operates two production lines for recycled-resin wall panels. During one quarter, quality engineers take independent simple random samples from the panels produced by each line. A sampled panel is classified as a first-pass success if it meets all thickness and surface standards without rework.

Line	Sample size	First-pass successes
A	180	151
B	170	132

Let p_A and p_B represent the true proportions of all panels produced during the quarter by Lines A and B, respectively, that would be first-pass successes. At the $\alpha = 0.05$ significance level, test the manufacturer's claim that the first-pass success proportion is higher for Line A than for Line B.

(a) State the hypotheses for the test.

(b) Check the conditions for the requested inference procedure. Assume each line produced more than 1,800 panels during the quarter.

(c) Calculate the appropriate test statistic and p-value. Show the quantities used in the calculation.

(d) State a conclusion in context using the $\alpha = 0.05$ significance level.

(e) Describe a Type II error for this test. If the true difference $p_A - p_B$ stays fixed and both sample sizes are substantially increased, explain the effect on the power of the test.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Defines the parameters in context as the quarter's true first-pass success proportions for all panels from Lines A and B.

- Point 2: States $H_0: p_A - p_B = 0$ and $H_a: p_A - p_B > 0$, or an equivalent pair of hypotheses.

(b) 2 points

- Point 1: Cites the independent SRSs and verifies the 10% condition because 180 and 170 are each no more than 10% of the corresponding line's quarterly production.

- Point 2: Checks the large-counts condition using the pooled proportion $\hat{p}_c = 283/350 \approx 0.8086$: expected successes and failures are approximately 145.54 and 34.46 for Line A and 137.46 and 32.54 for Line B, all at least 10.

(c) 3 points

- Point 1: Calculates $\hat{p}_A = 151/180 \approx 0.8389$, $\hat{p}_B = 132/170 \approx 0.7765$, $\hat{p}_A - \hat{p}_B \approx 0.0624$, and uses the pooled estimate $\hat{p}_c = 283/350 \approx 0.8086$.

- Point 2: Calculates $SE_{\text{pooled}} = \sqrt{[0.8086(1 - 0.8086)(1/180 + 1/170)]} \approx 0.04208$ and $z = 0.0624/0.04208 \approx 1.48$.

- Point 3: Calculates the one-sided p-value $P(Z \geq 1.48) \approx 0.069$.

(d) 1 point

- Point 1: Because $0.069 > 0.05$, fails to reject H_0 and concludes that the sample does not provide convincing evidence that Line A's quarter-wide first-pass success proportion is higher than Line B's.

(e) 2 points

- Point 1: Describes a Type II error as failing to find convincing evidence that $p_A > p_B$ when Line A's true first-pass success proportion actually is higher during the quarter.

- Point 2: Explains that increasing both sample sizes decreases the standard error, making a fixed true difference easier to detect and therefore increasing power.

Model Guidance:

The data involve two independent proportions, so use a two-proportion z test with the directional alternative $p_A > p_B$. Under H_0 , both samples estimate one common proportion; this is why the pooled estimate $283/350$ is used for the standard error. The random samples, the 10% checks, and all four pooled expected counts above 10 justify the normal approximation.

The observed difference is about 0.0624. Dividing by the pooled standard error of about 0.04208 gives $z \approx 1.48$, with a right-tail probability of about 0.069. This p-value exceeds 0.05, so the evidence does not meet the stated threshold. The conclusion is not that the two lines are proven equal; it is that this sample does not establish the claimed direction. A Type II error would occur if a real advantage for Line A were missed. Larger samples narrow the sampling distribution under a fixed alternative, raising the chance of rejecting H_0 when that advantage truly exists.

Expected Points:

10

APSTAT-FRQ1-004

Section: FRQ

Type: frq

Primary Unit: Unit 5, Regression Analysis

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Interpret Results, Integrated Analysis and Justification

Prompt:

A grocery-delivery company studied 12 daytime cargo-bike routes completed on dry days. For each route, x is the number of scheduled delivery stops and y is the percentage of a fully charged battery used during the route.

Route	A	B	C	D	E	F	G	H	I	J	K	L
Stops, x	8	9	10	11	12	13	14	15	16	18	20	22
Battery %, y	29	31	32	36	34	40	43	42	50	51	56	72

Computer output for all 12 routes:

$n = 12$

Least-squares line: $\hat{y} = 4.368 + 2.759x$

$r = 0.968$

$r^2 = 0.936$

$s = 3.313$ percentage points

When Route L is omitted, the least-squares line is $\hat{y} = 9.380 + 2.334x$ and $r = 0.980$.

(a) On paper, sketch a scatterplot of battery percentage used against number of stops. Label the axes, use appropriate scales, and describe the association, including any unusual feature.

(b) Interpret the slope 2.759 and $r^2 = 0.936$ in context.

(c) Use the line for all 12 routes to predict the battery percentage used by a route with 16 stops. Calculate and interpret the residual for Route I.

(d) An operations director states, “Adding 5 scheduled stops will cause every route to use about 14 additional percentage points of battery.” Use the regression output and the study design to evaluate this statement.

(e) Is Route L influential for the fitted regression relationship? Justify your answer by using its location in the data and the two fitted equations. Explain whether influence alone is a sufficient reason to delete Route L from the analysis.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Sketches the 12 ordered pairs with stops on the horizontal axis, battery percentage used on the vertical axis, clear labels, and scales covering at least $x = 8$ to 22 and $y = 29\%$ to 72%.
- Point 2: Describes a strong, positive, approximately linear association and notes that Route L at (22, 72) is far to the right and lies above the general pattern, making it a possible high-leverage or influential point.

(b) 2 points

- Point 1: Interprets the slope: among these dry-day daytime routes, each additional scheduled stop is associated with a predicted increase of about 2.759 percentage points in battery use, on average.
- Point 2: Interprets r^2 : about 93.6% of the variation in battery percentage used among these routes is accounted for by the linear relationship with number of scheduled stops.

(c) 2 points

- Point 1: Calculates $\hat{y} = 4.368 + 2.759(16) = 48.512\%$, and the residual for Route I as $y - \hat{y} = 50 - 48.512 = 1.488$ percentage points.
- Point 2: Interprets the positive residual: Route I used about 1.49 percentage points more battery than the model predicted for a 16-stop route.

(d) 2 points

- Point 1: Uses the slope to calculate an estimated increase of $5(2.759) = 13.795$, or about 13.8 percentage points, for a five-stop increase within the observed range, and recognizes this as an average model prediction rather than a guarantee for every route.
- Point 2: Rejects the causal wording because the routes were observed rather than randomly assigned a number of stops; other variables such as distance, hills, load, or rider could confound the relationship. Also limits use of the model to routes similar to the dry-day daytime routes with 8 to 22 stops.

(e) 2 points

- Point 1: Concludes that Route L is influential or potentially influential because its x -value is extreme and omitting it changes the slope from 2.759 to 2.334 and the intercept from 4.368 to 9.380, materially changing predictions and the estimated change per stop.
- Point 2: Explains that influence alone does not justify deletion; Route L should be removed only for a defensible reason such as a recording error or failure to belong to the target route conditions. If valid, it should be retained and the sensitivity of conclusions should be reported.

Model Guidance:

Plot the paired observations with stops as the explanatory variable. The points rise in a nearly linear pattern, and Route L is visually important because it is separated at the high end of x and sits above the trend of most routes. The fitted slope describes association, not causation. The r^2 value concerns variation in battery percentage accounted for by this linear model; it is not the percentage of routes predicted correctly.

At $x = 16$, the fitted value is 48.512%, so Route L's observed 50% produces a residual of about +1.49 percentage points. A five-stop change corresponds to 5 times the slope, or about 13.8 percentage points, but only as an average prediction for comparable routes within the observed range. Since stop counts were not assigned at random, the data cannot isolate a causal effect from route length, terrain, load, rider, or other variables.

Route L has leverage because its 22 stops place it at the edge of the x -distribution. Its influence is supported by the slope changing by about 0.425 percentage point per stop, from 2.759 to 2.334, when it is removed. That sensitivity should be disclosed, but a valid observation should not be discarded merely because it changes the model.

Expected Points:

10

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-FRQ1

- APSTAT-FRQ1-001: Unit 1 primary; no secondary unit; proportional stratified random sample by work shift followed by randomized blocks based on dispatcher experience; population of 240 regional emergency dispatchers and sample of 48; median correct ambulance-assignment time across 12 simulations; identify population/sample, justify stratification, formulate a comparative question, identify variables, design within-block random assignment, and distinguish causal scope from population generalizability; targets confusion between random sampling, random assignment, and blocking.
- APSTAT-FRQ1-002: Unit 2 primary; Unit 1 secondary; discrete distribution for the number of sediment-clogged filter cartridges during an 8-hour autumn coastal water-sampling shift, with $x = 0, 1, 2, 3, 4$ and probabilities 0.08, 0.27, 0.38, 0.20, 0.07; probability bar graph; validate probabilities, find $P(X \geq 3)$, calculate and interpret $\mu \approx 1.91$ and $\sigma \approx 1.03$, then calculate and interpret $P(X \geq 3 \mid X \geq 1) \approx 0.293$; targets treating a long-run noninteger mean as an impossible outcome and using the unconditional denominator for conditional probability.
- APSTAT-FRQ1-003: Unit 3 primary; Unit 2 secondary; one-sided two-proportion z test comparing quarterly first-pass success proportions for recycled-resin wall panels from independent SRSs on production Lines A and B; table gives 151 of 180 and 132 of 170 successes; define parameters, state $H_0: p_A - p_B = 0$ and $H_a: p_A - p_B > 0$, verify random/10%/pooled large-count conditions, pool to 283/350, calculate $z \approx 1.48$ and $p \approx 0.069$, fail to reject at $\alpha = 0.05$, describe Type II error and larger-sample effect on power; targets “fail to reject means equal” and failure to pool under H_0 .
- APSTAT-FRQ1-004: Unit 5 primary; Unit 1 secondary; paired table for 12 observed dry-day daytime cargo-bike routes using scheduled stops and battery percentage; scatterplot plus regression output $\hat{y} = 4.368 + 2.759x$, $r = 0.968$, $r^2 = 0.936$, $s = 3.313$; calculate prediction and residual at $x = 16$, interpret slope and r^2 , evaluate a five-stop causal/guaranteed claim, identify high-leverage Route L at (22, 72) as influential by comparing the full line with $\hat{y} = 9.380 + 2.334x$ after omission, and decide whether deletion is warranted; targets causal interpretation of observational regression, r^2 misinterpretation, extrapolation, and automatic removal of influential observations.